

Three-Mode Node-Error Comparison — Staged A→B Transmission

Generated 2026-07-10 16:35 UTC. Task: send the word "Quantum networks will link distant processors into one machine. This note crossed fifty kilometres of simulated fibre between two real quantum chips." (1192 bits, basis encoding) from node A to node B over a 50 km simulated fiber link, 256 shots per bit, majority-vote decoding.

Modes compared

- 1. REAL IBM HARDWARE — both node stages executed as circuits on the physical backend `ibm_marrakesh` (jobs: `d98hqmsqp3as739svodg`, `d98hqv0tcv6s73dm5gkg`).
- 2. DIGITAL TWIN — node stages on a local Aer simulator whose noise model is fitted from collected IBM calibration history (source: `fitted_model`).
- 3. EQUATIONS — node stages on Aer with every error rate derived in closed form from device physics (no measured data at all).

Control variables

The pipeline, message, shots, batching, link channel, and decoder are IDENTICAL in all three runs. Stage 2 (the link) is computed analytically from the composed channel map, so it is numerically identical across modes: every difference in the results is attributable to the node-error source alone.

Executive summary

Metric	Hardware	Twin	Equations
Received text	"Quantum networks will link distant processors into one machine. This n		
Message match	YES	YES	YES
Node-A raw BER	0.0081	0.0122	0.0108
Post-link BER (exp.)	0.1992	0.2017	0.2008
End-to-end raw BER	0.2027	0.2099	0.2084
Transmission fidelity	0.7973	0.7901	0.7916
Decoded bit errors	0/1192	0/1192	0/1192
Character accuracy	100.00%	100.00%	100.00%

Methodology — the measure-and-reprepare chain

Pipeline

Stage 1 (node A): each bit is encoded on a qubit (X gate for 1) and measured. On hardware this is a real batched SamplerV2 job; in twin/equations mode the same circuits run on Aer under the mode's node noise model. The measured per-bit $P(1)$ captures node A's gate + readout errors.

Stage 2 (link): the per-bit distribution is pushed through the 50 km link channel: all seven built-in fiber error channels composed into one CPTP map, from which the classical transition probabilities are computed exactly:

$$P(0 \rightarrow 1) = 0.06502 \quad P(1 \rightarrow 0) = 0.33161$$

The strong asymmetry comes from fiber attenuation modelled as amplitude damping: $|1\rangle$ decays toward $|0\rangle$, $|0\rangle$ is stable.

Stage 3 (node B): each bit's post-link $P(1)$ is re-prepared with $R_y(2 \cdot \arcsin(\sqrt{p}))|0\rangle$ — reproducing the Bernoulli distribution exactly — then measured, adding node B's errors. Majority vote over the shots decodes each bit.

Why not a single coherent circuit?

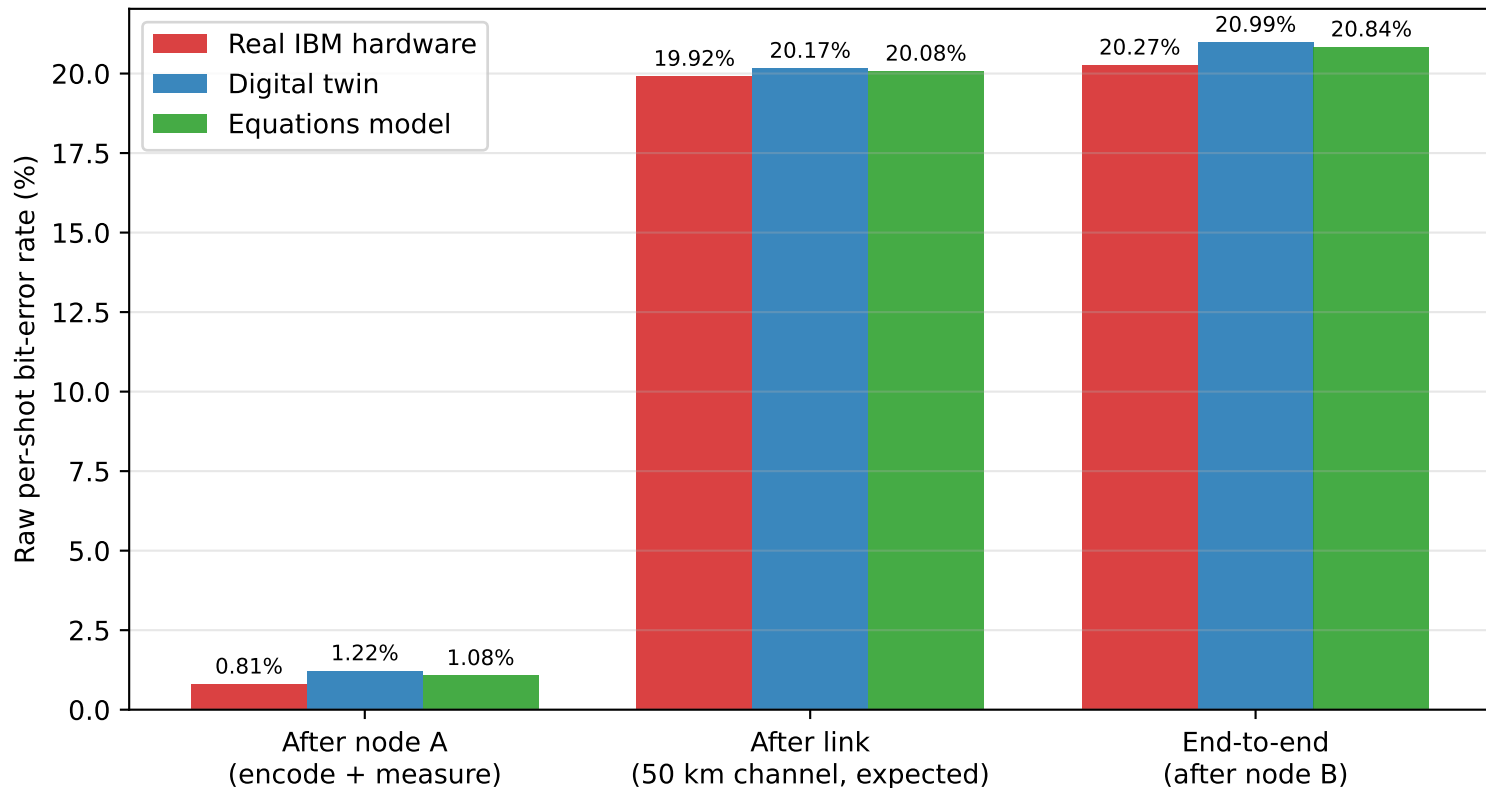
A cloud QPU accepts only circuits starting from $|0\dots 0\rangle$ and returns only classical counts: a quantum state can neither leave node A's chip nor enter node B's. Measure-and-reprepare at the node/link boundaries (a 'trusted-node relay', as deployed long-haul QKD networks use) is the only physically realizable hardware/simulation hybrid. For basis-encoded data the staged composition is statistically exact, because the post-measurement state entering the link is exactly $|0\rangle$ or $|1\rangle$; only coherence across stage boundaries is lost, which basis encoding never uses. Running the twin/equations modes through the SAME staged pipeline keeps the comparison apples-to-apples.

Decoding

Each bit is decided by majority vote over 256 shots; the raw per-shot BER (before the vote) is the informative error metric, since the vote corrects any per-shot error below 50%.

'Transmission fidelity' below is $1 - \text{raw end-to-end BER}$.

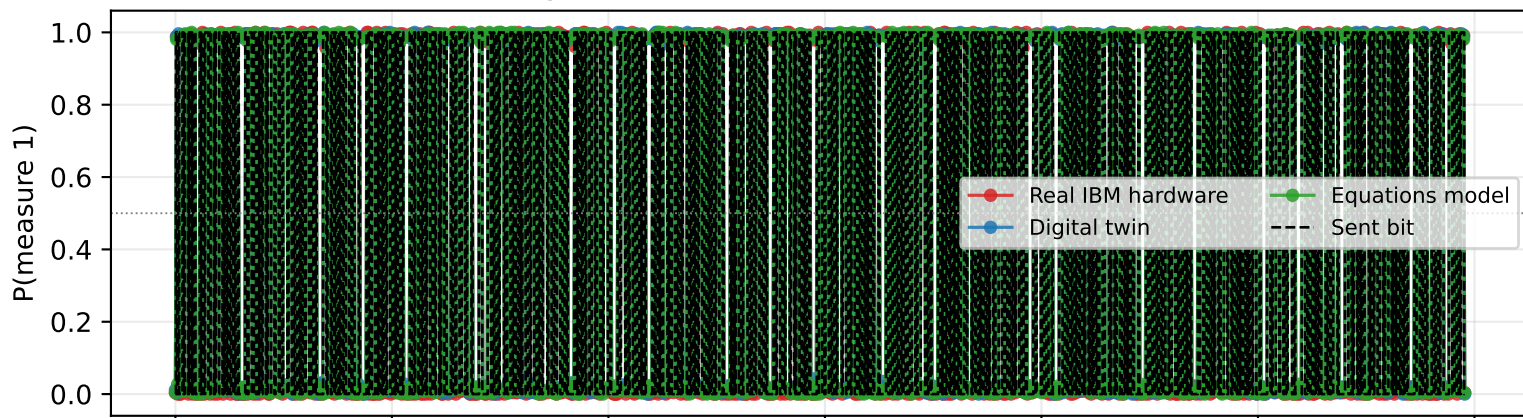
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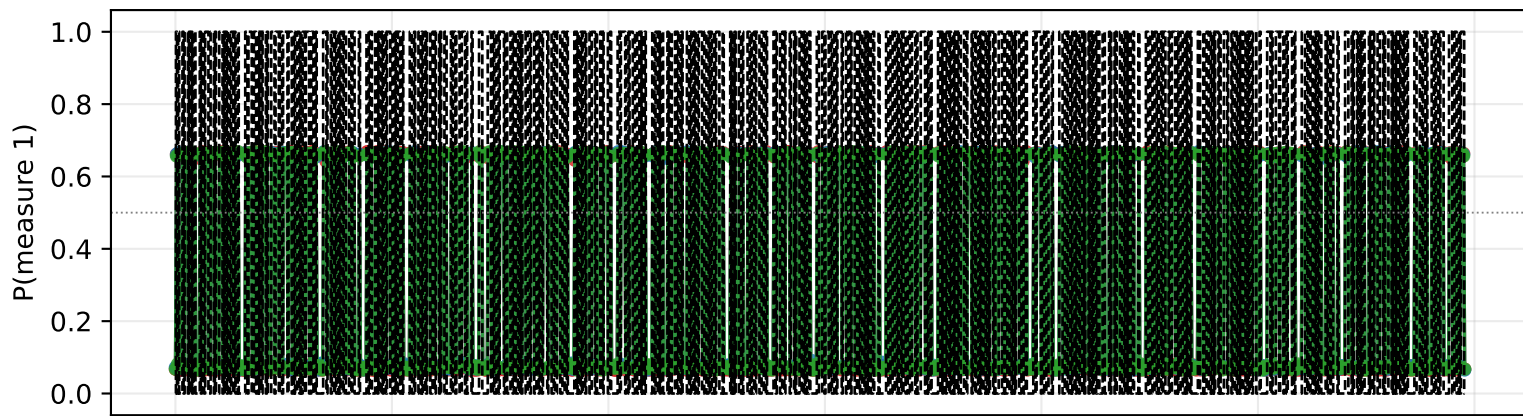
Raw per-shot BER after each pipeline stage. The link stage value is the exact expectation.

Per-bit P(1) through the pipeline — majority vote decodes each bit toward 0 or 1

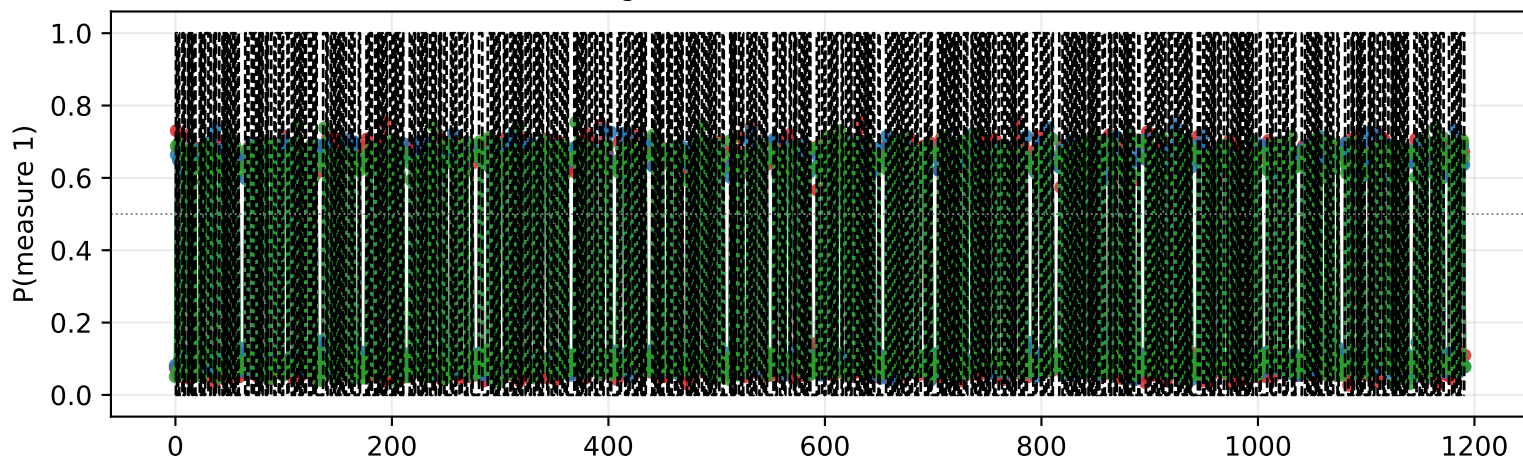
Stage 1 — after node A (measured on the node)



Stage 2 — after the simulated 50 km link

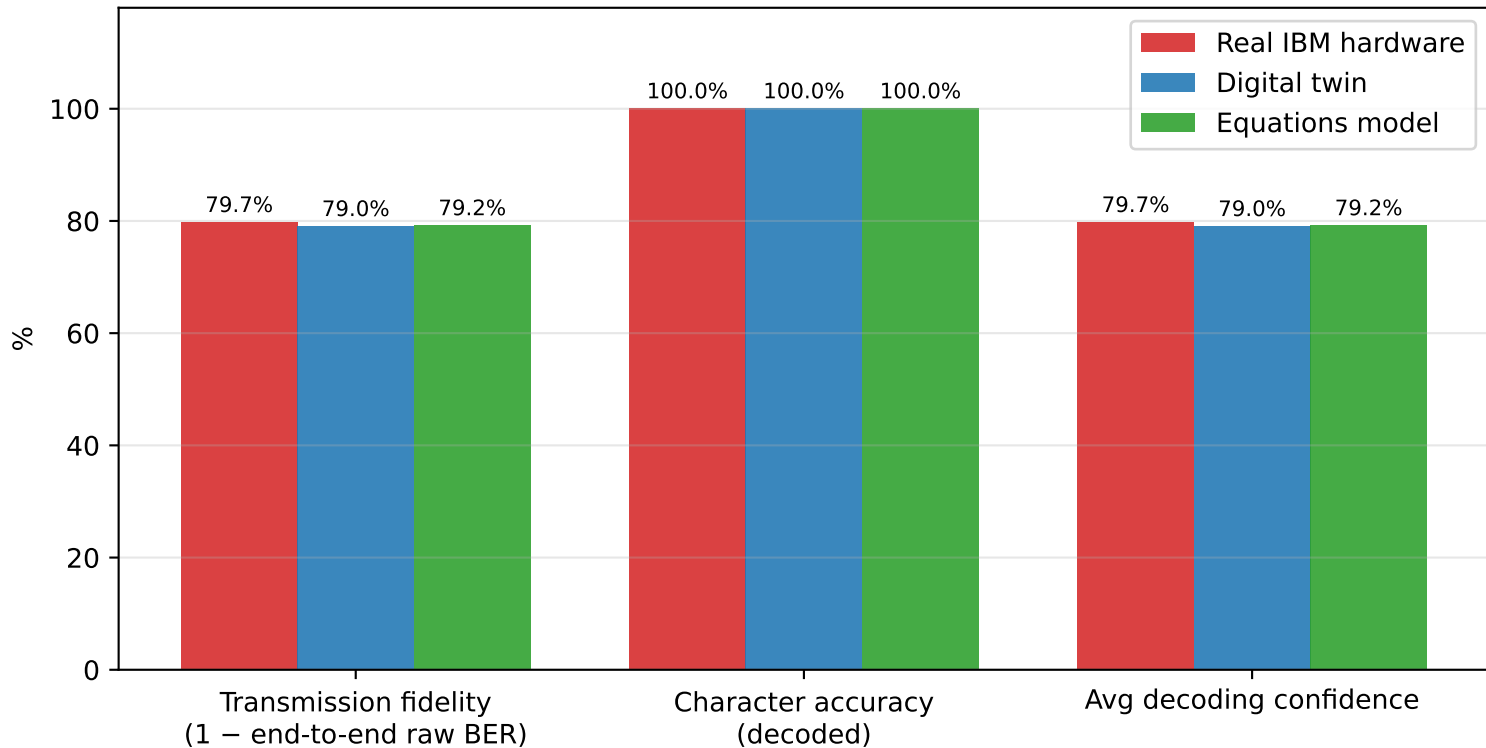


Stage 3 — after node B (end-to-end)

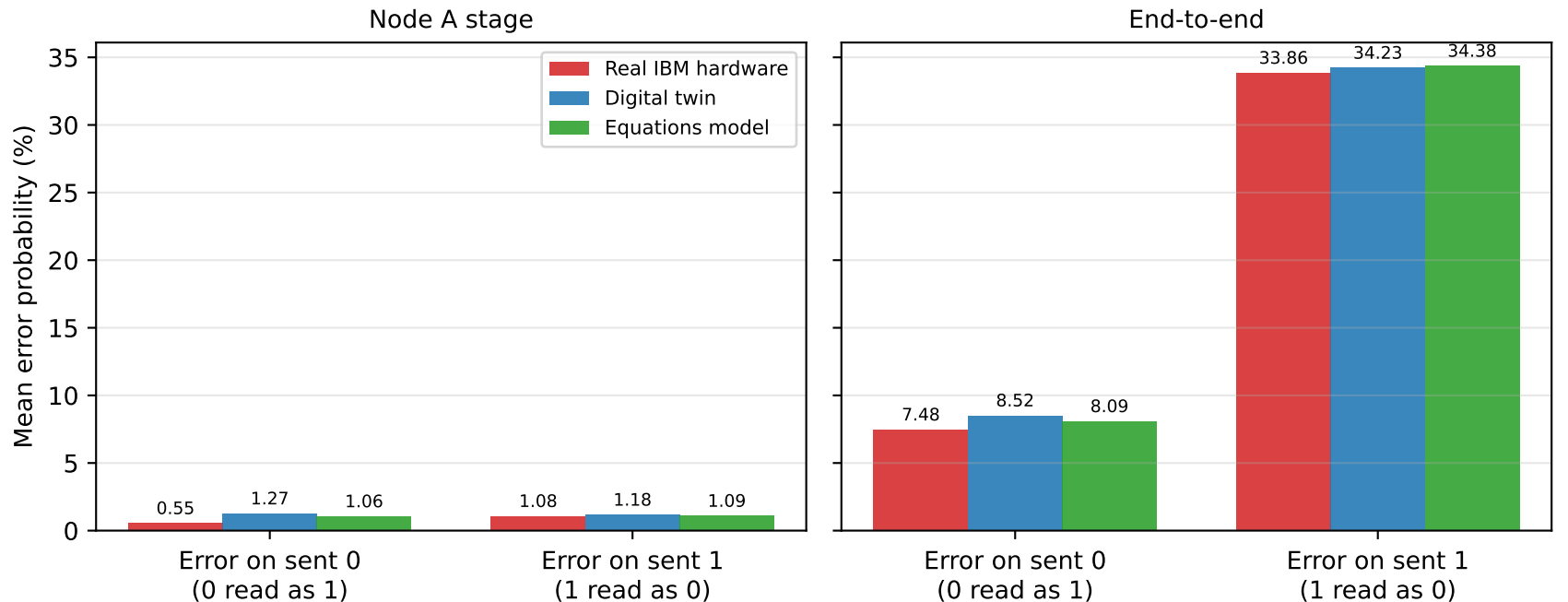


010110111001100101001011100010000010101000110100011010010111001100100000110111001101111011
Per-bit P(1) at each stage. Dashed black = the sent bit; dotted grey = the majority-vote decision threshold.

Fidelity metrics per mode

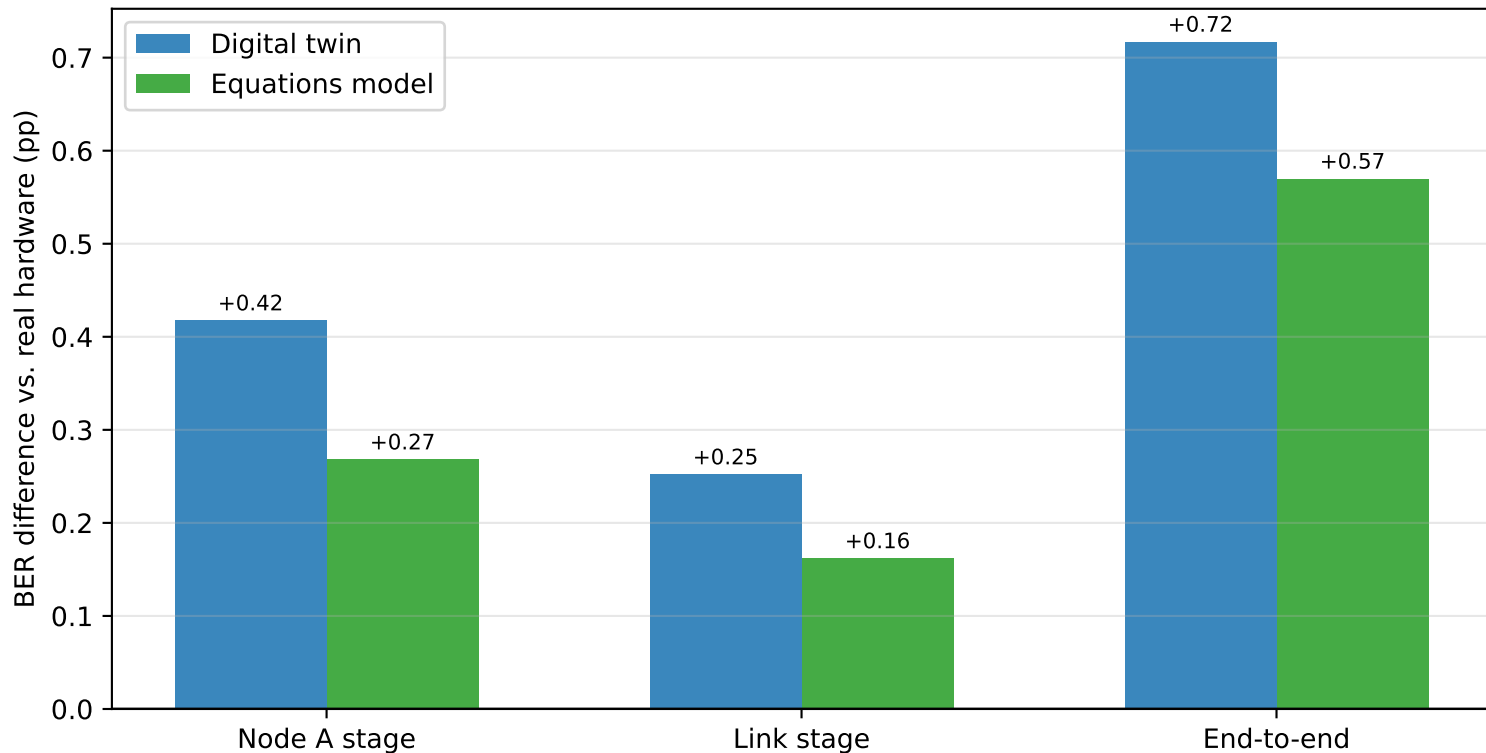


Error asymmetry: $|1\rangle$ decays toward $|0\rangle$ (T1 relaxation + amplitude damping), $|0\rangle$ is stable



Sent-0 vs sent-1 error rates. Real readout error is asymmetric; the twin uses the empirically fitted asymmetric readout matrix, the equations model symmetric readout.

Model discrepancy per stage (positive = model is more pessimistic than hardware)



Discrepancy analysis

Node-A stage (pure node error)

Hardware measured a raw BER of 0.0081 at the node-A stage; the twin predicted 0.0122 (1.52x of hardware) and the equations model 0.0108 (1.33x). This stage isolates encode-gate + readout error, so it is the cleanest test of the node-error models. The twin lands close to hardware, consistent with its prior validation (Bell fidelity 0.979 vs 0.990).

Link stage (identical channel, different inputs)

The link transition probabilities are identical for the three modes by construction. Small differences in post-link BER (see page 3) come only from each mode's different node-A output distribution being propagated through the same channel — i.e. they are inherited, not link discrepancies.

End-to-end

End-to-end raw BER: hardware 0.2027, twin 0.2099 (1.04x), equations 0.2084 (1.03x). The 50 km link dominates the budget in every mode (compare the node-A and post-link columns of the summary table): node errors are percent-level while the channel contributes tens of percent, mostly as 1->0 decay. All three modes agree on this structure, which is the key sanity check for using the free modes as stand-ins for hardware.

Structural discrepancies to expect

1. Readout asymmetry — real devices misread $|1\rangle$ more often than $|0\rangle$ (T1 relaxation during the readout window); the twin now applies an empirically fitted asymmetric readout matrix ($P(1|0)=0.0128$, $P(0|1)=0.0115$, fitted from earlier hardware runs — out-of-sample for this run); the equations model still symmetrizes its derived readout error. See the asymmetry chart (page 6).
2. Qubit selection — the transpiler places these small circuits on the device's better qubits, while the twin's median-over-functional-qubits rates represent a device-typical qubit: the twin is expected to be slightly pessimistic.
3. Drift — hardware calibration drifts between the twin's collected snapshots and the actual run.
4. Shot noise — with 256 shots per bit, one-sigma statistical uncertainty on a per-bit probability is up to ~ 3.1 pp, and stage-level BERs average over 1192 bits (305152 shots).

Verdict

All three modes decoded the message correctly — majority voting absorbs the channel's per-shot errors at this distance. The ordering of end-to-end BERs and the per-stage structure match between hardware and both free modes, supporting the simulator's use of twin/equations nodes for day-to-day work with hardware reserved for ground-truth validation.

Appendix — parameters, jobs, environment

Node calibrations used by the simulated modes

Parameter	Twin	Equations
T1 (us)	170.8	150.0
T2 (us)	105.0	100.0
1Q gate error	0.000355	0.000422
2Q gate error	0.002758	0.004663
Readout error	0.011041	0.010660

Twin calibration source

fitted_model for backend ibm_marrakesh.

Link channel

50 km, all seven built-in channels enabled (fiber attenuation, polarization drift, phase noise, timing jitter, detector dark counts, environmental decoherence, chromatic dispersion), composed into one Kraus channel; $P(0 \rightarrow 1) = 0.06502$, $P(1 \rightarrow 0) = 0.33161$.

IBM hardware jobs

Backend: ibm_marrakesh

Stage-1 job: d98hqmsqp3as739svodg

Stage-3 job: d98hqv0tcv6s73dm5gkg

Estimated quota consumed by this run: 565.9 s

Month-to-date usage: 40.0 s of 600 s (6.7%).

Software

python: 3.12.3

qiskit: 2.5.0

qiskit_aer: 0.17.2

numpy: 2.5.0

Data files (same folder as this report)

mode_hardware.json / mode_twin.json / mode_equations.json — full per-stage, per-bit data for each mode

comparison.json — side-by-side summary and discrepancy metrics

chart_*.png — every chart as a standalone image